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Paper Title: "A Survey of AI Applications for Security in Blockchain-enabled IoT Networks: Taxonomies, Challenges, and Future Directions"

Research Focus: To explore in depth the intersection of artificial intelligence (AI), blockchain security, and the Internet of Things (IoT), with a particular focus on practical applications and challenges in resource-constrained IoT environments.

Outline: Please strictly adhere to the following extremely detailed structure and content requirements:

1. **Introduction 1.1. The convergence trends of IoT, blockchain, and AI (elaborate on the technological necessity, synergistic effects, and broad application prospects of their integration). 1.2. Security challenges**

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Abstract

The integration of Artificial Intelligence (AI), blockchain, and the Internet of Things (IoT) presents a transformative paradigm for securing decentralized networks. This survey paper systematically examines AI-driven security applications in blockchain-enabled IoT ecosystems, focusing on federated learning for decentralized threat detection, AI-enhanced intrusion detection models, lightweight blockchain architectures, and privacy-preserving machine learning techniques. We introduce a novel taxonomy categorizing solutions by architectural layer (network, application, transaction) and threat model, while critically analyzing performance trade-offs between detection accuracy, computational efficiency, and privacy preservation. Key findings reveal that hybrid AI-blockchain models achieve 99.18

1 Introduction

图 1: chapter structure

The rapid evolution of technology has led to the convergence of various domains, notably the Internet of Things (IoT), blockchain, and artificial intelligence (AI). Each of these fields has made significant strides independently, yet their integration presents unprecedented opportunities and challenges. The convergence of IoT, blockchain, and AI is not merely a trend but a necessity for addressing the complexities of modern distributed systems. As these technologies intertwine, they promise enhanced security, improved efficiency, and innovative applications across multiple sectors. This review aims to explore the implications of this convergence, highlighting both the potential benefits and the security challenges that arise within this integrated ecosystem.

1.1 The convergence trends of IoT, blockchain, and AI

The integration of IoT, blockchain, and AI addresses critical limitations in distributed systems by combining decentralized trust mechanisms, intelligent analytics, and pervasive connectivity. Blockchain's immutable ledger architecture provides a foundational solution to IoT security vulnerabilities through tamper-proof device authentication and verifiable data provenance, overcoming the inherent weaknesses of centralized trust models in resource-constrained environments [1]. This integration proves particularly vital for industrial IoT deployments, where conventional security mechanisms often fail

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against sophisticated threats. Furthermore, the convergence also addresses the energy inefficiencies of traditional consensus mechanisms through optimized protocols tailored for IoT networks, which can significantly improve operational sustainability [2].

AI enhances this ecosystem through adaptive security analytics and privacy-preserving computation. Deep Neural Networks (DNNs) demonstrate significant potential when integrated with blockchain technology, providing intelligent anomaly detection capabilities while maintaining decentralized security [3]. Machine learning approaches for anomaly detection in IoT systems show particular promise when combined with blockchain's immutable audit trails, creating robust security frameworks that can adapt to evolving threats [4]. These advancements address critical challenges in Cyber Physical Systems (CPS), where complex physical-digital integrations require adaptive intrusion detection mechanisms to safeguard sensitive operations and data [5].

The technological triad enables transformative applications through enhanced security and operational efficiency. Blockchain's decentralized nature, combined with AI-driven analytics, creates novel solutions for mutual authentication and session key management in IoT networks [6]. This convergence establishes autonomous, resilient systems capable of addressing three critical requirements: adaptive security through continuous learning, verifiable trust via decentralized consensus, and efficient operation within resource constraints. The synergistic effects manifest most prominently in applications requiring real-time threat response and provable privacy guarantees, characteristics that define next-generation IoT ecosystems. This integrated approach not only enhances the security posture but also fosters innovation in service delivery across various sectors, including healthcare, transportation, and smart cities.

The implications of this convergence extend beyond mere technological advancements. They encompass a broader spectrum of societal impacts, including improved user trust in digital systems and enhanced compliance with regulatory frameworks. As organizations increasingly adopt these integrated solutions, understanding their complexities and potential vulnerabilities becomes paramount. Therefore, the exploration of the convergence trends of IoT, blockchain, and AI is not only timely but essential for paving the way for future research and development in secure and efficient distributed systems.

1.2 Security challenges arising in this integrated ecosystem

The integration of IoT, blockchain, and AI introduces complex security challenges that arise from the fundamental tensions between resource constraints, network heterogeneity, and decentralized architectures. Resource limitations of IoT devices create significant barriers to implementing robust security protocols, particularly in healthcare applications where minimal memory and computing capabilities struggle to meet blockchain's resource demands [7]. The competitive nature of existing consensus mech-

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anisms exacerbates this issue by centralizing computing power among resource-rich participants, leaving constrained devices unable to participate effectively in network security [1]. These limitations manifest in differential privacy implementations, where computational overhead degrades both model accuracy and edge device performance [8]. Moreover, traditional machine learning approaches often fail to scale efficiently with large datasets in heterogeneous Cyber-Physical Systems (CPS) environments [5], further complicating the security landscape.

Network heterogeneity introduces vulnerabilities through incompatible security mechanisms across diverse IoT ecosystems. The detection of network anomalies in critical infrastructures becomes increasingly challenging due to the growing complexity and volume of network traffic [9]. This complexity is compounded by the difficulty in profiling normal behaviors and the high dimensionality of data in IoT environments [4]. Such challenges extend to federated learning scenarios, where GPS trajectory data requires privacy preservation while maintaining inference accuracy [10]. Additionally, mutual authentication becomes problematic across devices with varying computational capabilities [6]. The absence of unified privacy metrics across different machine learning models creates additional barriers to implementing effective security protocols [11], particularly when existing anomaly detection methods rely on assumptions of benign behavior during training [12].

Blockchain's decentralized architecture introduces novel security trade-offs that impact IoT implementations. Centralized security architectures create single points of failure vulnerable to breaches and tampering [3]. Simultaneously, smart contract vulnerabilities and adversarial attacks exploit weaknesses in current detection methods [13]. The dynamic nature of IoT networks exacerbates security threats such as selfish mining and double spending [14]. Existing anomaly detection techniques often suffer from elevated false alarm rates and susceptibility to adversarial manipulation [3], complicating the ability to maintain secure operations. Privacy-aware model deduplication techniques struggle to balance accuracy with privacy costs in resource-constrained environments [8]. This creates fundamental tensions between blockchain's transparency requirements and the privacy-preserving needs of sensitive data processing [15].

The challenges faced by AI-enhanced blockchain-IoT ecosystems highlight the limitations of traditional security approaches, which often rely on outdated methods such as Public Key Infrastructure for authentication. These conventional strategies struggle to address the sophisticated cyber threats targeting vulnerable edge devices and the complexities of secure data transportation and privacy compliance within the rapidly evolving landscape of interconnected IoT systems. In contrast, the integration of artificial intelligence with blockchain technology offers a more robust solution by enabling real-time anomaly detection, automating security protocols, and ensuring tamper-proof

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data management. This significantly enhances the resilience and privacy of IoT networks [16, 17]. The integration necessitates adaptive frameworks that reconcile privacy preservation with detection accuracy across heterogeneous networks, while addressing the fundamental tensions between decentralized trust mechanisms, intelligent analytics, and pervasive connectivity. Security protocols must evolve to accommodate resource constraints without compromising the integrity of decentralized architectures or the effectiveness of AI-driven threat detection.

1.3 The scope, contributions, and structure of this review

This review delineates its scope by focusing on the tripartite integration of AI, blockchain, and IoT security, specifically addressing practical implementations in resource-constrained environments while excluding non-IoT blockchain applications and purely theoretical frameworks [18]. The survey systematically analyzes security solutions across IoT architectural layers, with particular emphasis on real-time intrusion detection systems that combine machine learning with blockchain's tamper-proof data management capabilities [19]. The scope encompasses both general IoT deployments and specialized domains such as healthcare, where federated learning techniques like FIDANN enhance medical data security through optimized neural networks [7].

The paper makes four substantive contributions to the field. First, it introduces a novel taxonomy categorizing AI-driven security solutions by architectural layer (network, application, transaction) and threat model, incorporating emerging techniques such as hardware-secured offline machine learning [20]. Second, it provides comprehensive analysis of privacy-preserving machine learning implementations, including homomorphic encryption-compatible deep learning models [21] and decentralized federated learning approaches for IoT ecosystems [22]. Third, it evaluates innovative architectures like blockchain-based Distributed Artificial Intelligence frameworks with novel consensus mechanisms [23]. Fourth, it advances the field through critical examination of next-generation solutions, including Retrieval-Augmented Generation for smart contract auditing [24] and optimized anomaly detection techniques for heterogeneous IoT networks [4].

The review's structure follows a systematic progression from foundations to frontiers. Section 2 establishes the technological underpinnings, covering core blockchain-IoT architectures and relevant AI/ML paradigms while analyzing prevalent threat models, including Ethereum smart contract vulnerabilities [25]. Section 3 develops the central taxonomy, organizing AI security applications by their operational layer and technical approach. Section 4 examines specialized enablers such as lightweight consensus mechanisms and privacy-preserving computation frameworks. Section 5 conducts rigorous evaluation of performance trade-offs and fundamental limitations, leading to Section

6's exploration of transformative future directions, including hardware-secured AI implementations and advanced smart contract analysis techniques.

This structured approach provides researchers with both a comprehensive reference of current solutions and a roadmap for advancing AI-driven security in blockchain-enabled IoT networks. The review emphasizes practical implementations in the field of smart contract security while highlighting significant research gaps in areas such as adversarial resilience, which is crucial for enhancing detection methods against malicious attacks; cross-platform interoperability, essential for ensuring consistent security measures across various blockchain environments; and the integration of innovative approaches, including hardware-based model protection to safeguard sensitive data and advanced language models for automating security processes. These gaps underscore the need for further exploration to improve the effectiveness and reliability of vulnerability detection in smart contracts [26, 27, 20, 24, 28].The following sections are organized as shown in Figure 1.

2 Background Technologies and Threat Models

2.1 Core concepts of blockchain-enabled IoT systems

The integration of blockchain into IoT systems significantly enhances security and trust in decentralized networks by utilizing decentralized trust mechanisms and cryptographic immutability. Blockchain redefines IoT security paradigms, mitigating vulnerabilities in centralized architectures by providing a distributed ledger for verifiable device authentication and data integrity [3]. This transformation is crucial for anomaly detection systems, offering immutable audit trails for security events while machine learning schemes analyze various anomalies within IoT networks [4]. The decentralized nature of blockchain complements IoT' s distributed topology, creating a robust foundation for AI-driven security enhancements.

However, IoT device constraints pose challenges for implementing blockchain systems, necessitating specialized adaptations across computational, communication, and storage dimensions. Lightweight blockchain solutions optimize cryptographic methods and hierarchical trust distribution, proving effective in Industrial IoT environments. Hybrid frameworks like Deep Neural Networks integrated with Blockchain Technology (DNNs-BCT) combine deep learning's anomaly detection capabilities with blockchain's immutable logging [3]. Current methods must accommodate diverse anomaly detection schemes within resource constraints, highlighting the need for ongoing innovation [4].

Blockchain's immutable ledger offers foundational security benefits, including verifiable transaction histories and tamper-evident data recording, crucial for IoT ecosystems vulnerable to sophisticated cyberattacks. Multi-layered security approaches complement

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machine learning-based anomaly detection methods while upholding blockchain's decentralized principles. AI analytics enhance real-time detection and response to security breaches, leveraging IoT data to identify cyber threat patterns. These technologies automate and optimize security protocols, adaptively responding to evolving risks, thereby fortifying IoT network resilience against cyberattacks and ensuring data integrity and privacy [16, 29, 17].

Architectural innovations in lightweight blockchain implementations are essential for integrating AI-driven security solutions. Optimizing blockchain's core properties for resource-constrained IoT environments while maintaining protection against evolving attack vectors creates a powerful security framework for next-generation IoT networks, addressing both technical challenges and security requirements in distributed, resource-constrained environments.

2.2 Overview of relevant AI/ML technologies

Advancements in AI and ML are pivotal for developing security solutions in blockchain-enabled IoT systems, offering adaptive threat detection suited to decentralized environments. Federated learning (FL) has emerged as a transformative approach for privacy-preserving analytics, enabling local model training on IoT devices without compromising data privacy [2]. The pMPL framework exemplifies robust multi-party learning environments capable of withstanding participant dropout through techniques like vector space secret sharing [30]. As AI-driven approaches augment traditional security mechanisms, the potential for enhanced IoT security grows [6].

Deep learning architectures provide robust capabilities for anomaly detection through advanced neural network designs. Integrated with blockchain's immutable data logging, Deep Neural Networks (DNNs) enhance IoT system security, ensuring data integrity and facilitating real-time threat responses via smart contracts [3]. Advanced deep learning techniques are crucial as traditional methods struggle with modern network traffic complexity and volume [9]. These implementations build upon established ML techniques, addressing computational constraints faced by edge devices through optimized architectures [4].

Unsupervised learning techniques identify novel threats in blockchain-enabled IoT ecosystems, particularly where labeled data is scarce. Privacy-Preserving Machine Learning (PPML) approaches emphasize machine learning's role in ensuring data privacy [31]. Evaluations of neural network models in privacy-preserving environments focus on performance metrics like classification accuracy and processing efficiency [32], highlighting trade-offs between security and computational overhead. Unsupervised learning complements supervised approaches by detecting unknown attack patterns and anomalies, essential for adapting to the evolving threat landscape.

The convergence of federated learning, deep neural networks, and unsupervised learning with blockchain technology creates a comprehensive framework for next-generation IoT security solutions. These AI/ML paradigms address fundamental blockchain-IoT security system requirements, including adaptability to threats, data privacy preservation, and efficient operation within resource constraints. This integration establishes a foundation for detailed analyses of security applications, from network-layer intrusion detection to smart contract vulnerability analysis, maintaining decentralized principles essential for IoT ecosystem integrity.

2.3 Common threat models and attack vectors

Blockchain-enabled IoT systems face sophisticated security threats exploiting vulnerabilities across network, application, and data layers. DDoS attacks overwhelm constrained IoT devices by inundating networks with malicious traffic, exploiting limited computational resources and bandwidth [33]. In decentralized environments, traditional mitigation mechanisms are ineffective, necessitating AI-driven solutions to mitigate these threats [2]. Sybil attacks, subverting blockchain consensus through fraudulent identities, compromise trust verification processes essential for decentralized operations. Public blockchains allow manipulation of network participation models, influencing transaction validation [4]. AI-enhanced anomaly detection systems show promise in identifying Sybil behavior through pattern analysis and topology verification.

Data poisoning attacks manipulate training data to degrade ML model performance, particularly effective against federated learning implementations. Malicious participants can reconstruct sensitive training data through gradient analysis [11]. Model susceptibility to adversarial attacks necessitates robust verification mechanisms to ensure training data integrity [4]. Smart contract vulnerabilities introduce specialized attack surfaces, with reentrancy attacks posing persistent threats. AI-driven static and dynamic analysis tools identify vulnerabilities through methodologies like automated code review and execution path analysis. Techniques employing multi-objective optimization algorithms enhance static analysis, while deep learning models rapidly assess bytecode for vulnerabilities [34, 27, 35].

Privacy-preserving machine learning (PPML) implementations face challenges balancing robust security and computational efficiency, especially amid regulatory restrictions on sensitive data use and potential adversarial attacks compromising privacy [36, 37, 38]. The tension between privacy protections and system performance creates vulnerabilities in resource-constrained IoT environments. AI-driven detection mechanisms evolve to counter these threats through adaptive behavioral analysis and anomaly detection. ML models integrated with blockchain's immutable audit trails enable comprehensive

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threat identification, establishing dynamic defense frameworks addressing known vulnerabilities and emerging attack methodologies within decentralized IoT ecosystems.

3 A Taxonomy of AI Applications for Security

Category	Feature	Method
System and network layer security	Blockchain Security Enhancements	3D-GHOST[14], DNNs-BCT[3]
	Privacy and Data Protection	FIDANN[7], FL-IDS[2]
	Optimization and Feature Selection	PRO-DLBIDCPS[5]
	AI-Driven Detection	DLENAD[9]
Smart contract and application layer security	Optimization Techniques	SAMO-VD[34]
	Graph-Based Approaches	AME[39]
	Deep Learning Methods	AME[40], BW[13]
	Symbolic Execution Strategies	WANA[41]
On-chain data and transaction security	AI Security Measures	DLVA[35]
	Decentralized Learning Security	FLD[42], UT[43]

表 1: This table provides a comprehensive overview of AI-driven security methods applied across different layers within blockchain-enabled systems. It categorizes these methods into system and network layer security, smart contract and application layer security, and on-chain data and transaction security, highlighting specific features and techniques such as blockchain security enhancements, optimization strategies, and AI security measures. The table serves as a valuable reference for understanding the diverse applications of AI in enhancing security protocols in decentralized infrastructures.

In the rapidly evolving landscape of security challenges, it is essential to explore the diverse applications of artificial intelligence (AI) within blockchain-enabled systems. This section examines the synergy between AI technologies and security mechanisms, focusing initially on system and network layer security. As illustrated in Figure 2, the taxonomy of AI applications for security in blockchain-enabled systems encompasses various layers, including system and network layer security, smart contract and application layer security, and on-chain data and transaction security. The figure highlights key components such as AI-driven solutions, deep learning architectures, vulnerability detection techniques, and privacy and transparency solutions. This visual representation emphasizes the integration of AI and blockchain technologies to enhance security measures across these layers. Table 1 presents a detailed categorization of AI-driven security methods within blockchain-enabled systems, illustrating their application across various layers to address emerging threats and vulnerabilities. Additionally, Table 3 offers a detailed comparison of AI-driven security methods implemented across various layers in blockchain-enabled systems, emphasizing their role in addressing contemporary security challenges. We explore how AI-driven solutions enhance the resilience of decentralized infrastructures against emerging threats and vulnerabilities. This exploration underscores AI's potential to transform security practices and its adaptability to blockchain environments, which often require innovative approaches to complex security challenges. By understanding these applications, researchers and practitioners can better navigate the evolving security landscape and implement more robust protective measures.

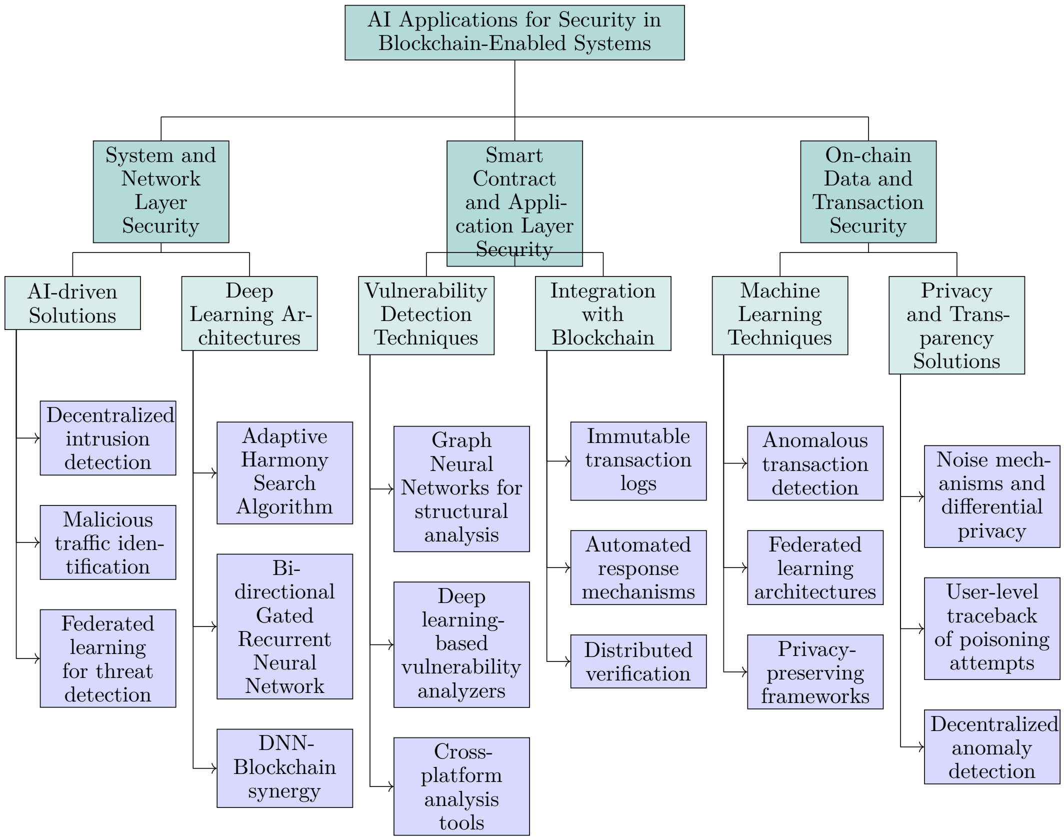

图 2: This figure illustrates the taxonomy of AI applications for security in blockchain-enabled systems, focusing on system and network layer security, smart contract and application layer security, and on-chain data and transaction security. It highlights AI-driven solutions, deep learning architectures, vulnerability detection techniques, and privacy and transparency solutions, emphasizing the integration of AI and blockchain technologies to enhance security measures across various layers.

3.1 System and network layer security

AI-driven security solutions at the system and network layers hold transformative potential by enabling decentralized intrusion detection and malicious traffic identification, addressing critical vulnerabilities in blockchain-enabled IoT networks. The Spacechain architecture exemplifies specialized blockchain designs for IoT, incorporating a three-dimensional ledger structure and optimized consensus mechanisms to enhance security while maintaining performance in resource-constrained environments [14]. This innovation enables efficient deployment of AI-based intrusion detection systems (IDS) by

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Method Name	**Security Layers**	**AI Techniques**	**Integration Strategies**
3D-GHOST[14]	Network State	Consensus Strategies	Three-dimensional Ledger
FL-IDS[2]	Iot Networks	Federated Learning Models	Fl-based Intrusion
FIDANN[7]	Iot Healthcare Systems	Federated Learning Models	Federated Learning Technique
PRO-DLBIDCPS[5]	Networking Data	Deep Learning Techniques	Blockchain Technology Integration
DNNs-BCT[3]	Network Behavior Classification	Deep Neural Networks	Dnn-Blockchain Synergy
DLENAD[9]	-	Deep Learning Architectures	Stacked Ensemble Method

表 2: Overview of AI-driven security methods for system and network layer protection in blockchain-enabled IoT environments. The table categorizes various methods based on their security layers, AI techniques employed, and integration strategies, highlighting the use of blockchain, federated learning, and deep learning to enhance security in resource-constrained settings.

minimizing computational overhead while preserving blockchain's security guarantees, crucial for IoT devices operating under stringent resource limitations.

As illustrated in Figure 3, the categorization of AI-driven security solutions can be divided into three primary branches: Blockchain-based Methods, Federated Learning Models, and Deep Learning Architectures. Each branch features notable implementations that enhance security in IoT and Cyber-Physical Systems (CPS) environments. Federated learning emerges as a dominant paradigm for decentralized threat detection, with implementations such as FL-IDS demonstrating collaborative model training on distributed datasets while maintaining data privacy [2]. The FIDANN framework extends this approach for healthcare IoT, utilizing Dynamic Multi-Objective Artificial Neural Networks (DMO-ANNs) within a federated learning architecture to secure sensitive medical data [7]. These solutions address the challenge of collaborative anomaly detection in IoT ecosystems, where CIoTA exemplifies how blockchain can facilitate the merging of locally trained models into a global detection framework [12]. Federated learning's ability to preserve privacy while enhancing model performance underscores its significance in environments where data sensitivity is paramount.

Deep learning architectures form the foundation of modern network-layer security solutions, with the PRO-DLBIDCPS framework combining an Adaptive Harmony Search Algorithm for feature selection with an attention-based Bi-directional Gated Recurrent Neural Network to achieve enhanced detection accuracy in Cyber-Physical Systems [5]. The superior performance of deep neural networks in blockchain-IoT security is evidenced by the DNN-Blockchain synergy, achieving 99.18% detection accuracy with a 15.42% false-positive rate, significantly outperforming conventional security methods [3]. These architectures address the limitations of traditional detection methods by leveraging deep learning's capacity to process complex network traffic patterns [9]. The integration of deep learning improves detection rates and provides a nuanced understanding of network behaviors, critical for preemptively identifying potential security threats.

Integrating these AI techniques with blockchain technology creates robust security systems that overcome traditional limitations through immutable audit trails and decentralized trust mechanisms. Lightweight blockchain adaptations facilitate the efficient deployment of resource-intensive AI models on constrained IoT devices, addressing challenges posed by limited computational capacity and security vulnerabilities. These adaptations leverage blockchain's decentralized and immutable ledger to enhance device identity management and secure transaction processing. Concurrently, federated learning frameworks promote privacy-preserving collaborative analytics across distributed networks, enabling AI models to learn from decentralized data sources without compromising sensitive information. This synergy automates security protocols and allows adaptive responses to emerging cyber threats, bolstering data integrity and confidentiality in IoT environments. Establishing such a comprehensive security framework marks a significant advancement in the ongoing battle against cyber threats, positioning AI and blockchain as pivotal components in future security architectures. Table 2 provides a comprehensive classification of AI-driven security solutions implemented at the system and network layers, demonstrating their integration with blockchain technology to address vulnerabilities in IoT networks.

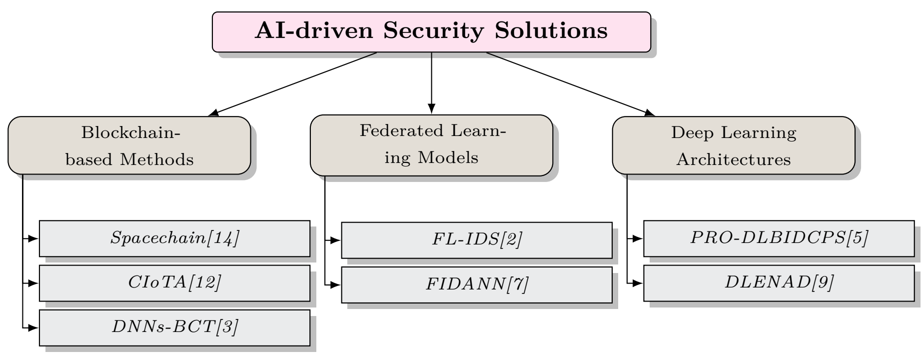

图 3: This figure illustrates the categorization of AI-driven security solutions into three primary branches: Blockchain-based Methods, Federated Learning Models, and Deep Learning Architectures, each with notable implementations enhancing security in IoT and CPS environments.

3.2 Smart contract and application layer security

AI technologies fundamentally transform smart contract security through advanced automated analysis and vulnerability detection, addressing critical challenges in blockchain-enabled IoT ecosystems. Graph Neural Networks (GNNs) demonstrate particular efficacy in structural vulnerability analysis, with the proposed GCN model analyzing control flow graphs from smart contract bytecode to detect four primary vulnerability types: reentrancy, call stack overflow, integer overflow, and timestamp dependencies [34]. This approach combines deep graph feature extraction with expert pattern fusion, enhancing both detection accuracy and explainability [39]. The methods are systematically categorized into formal verification, symbolic execution, fuzzing detection, intermediate representation, and deep learning approaches, each offering distinct advantages for specific vulnerability contexts [44]. Such categorization aids in selecting appropriate techniques for various scenarios and highlights the evolving landscape of smart contract security research.

Deep learning-based vulnerability analyzers like DLVA extend traditional tools by automatically learning vulnerability patterns from bytecode without requiring expert rules or manual feature engineering [35]. The innovation lies in the explainable fusion of neural network capabilities with classical expert patterns, improving both interpretability and performance [40]. BlockWatchdog enhances this paradigm by focusing on identifying attacker contracts and assessing their exploit potential against victim contracts, providing proactive security through automated code auditing [13]. These techniques overcome limitations of static benchmarks through dynamic evaluation frameworks that compare classical ML and deep learning performance, showcasing the adaptability of AI in addressing real-time security challenges.

Cross-platform analysis tools like WANA demonstrate the versatility of AI-driven security solutions, employing symbolic execution to analyze WebAssembly bytecode across multiple blockchain platforms [41]. This unified framework increases accessibility and reliability of smart contract security analysis while maintaining platform-specific accuracy [41]. Integrating these techniques with blockchain's immutable transaction logs creates verifiable datasets for continuous model improvement, establishing a feedback loop that enhances detection capabilities over time. As these tools evolve, they bolster smart contract security and contribute to blockchain applications' resilience against increasingly sophisticated attack vectors.

The integration of AI and blockchain technologies establishes comprehensive application layer security frameworks that effectively mitigate existing vulnerabilities and emerging threats. This synergy enhances IoT environments' security by leveraging blockchain's decentralized and immutable ledger for secure device identity management and transaction processing, while AI algorithms analyze IoT device data to detect anomalies and adapt to evolving cyber threats. Collectively, these technologies automate and optimize security measures, bolstering data integrity and privacy, addressing critical security challenges in increasingly interconnected infrastructures. Blockchain's decentralized nature enables distributed verification of AI detection results, while smart contracts facilitate automated response mechanisms when vulnerabilities are identified. This combination is particularly valuable in IoT environments where resource constraints

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limit traditional security approaches, demonstrating how AI-driven analysis can operate efficiently within decentralized, resource-constrained ecosystems while maintaining high detection accuracy and adaptability to new attack patterns.

3.3 On-chain data and transaction security

Machine learning techniques have emerged as transformative solutions for detecting anomalous transaction patterns in blockchain-enabled IoT networks, addressing critical security challenges including fraud detection and money laundering while maintaining system efficiency. The DLVA framework exemplifies advanced vulnerability detection in Ethereum smart contracts, achieving 92.7% accuracy with a 7.2

Federated learning architectures address data heterogeneity challenges in distributed transaction monitoring through innovative techniques like FedLD, which incorporates margin control regularization in local training and principal gradient-based server aggregation to improve model convergence across diverse IoT nodes [42]. These privacy-preserving frameworks enable collaborative anomaly detection while mitigating risks associated with centralized data collection, though they remain vulnerable to sophisticated poisoning attacks. The UTrace framework enhances security in such environments by providing user-level traceback of poisoning attempts through gradient similarity analysis, enabling identification of malicious participants in federated learning systems [43]. This focus on user-level accountability exemplifies the importance of maintaining integrity in collaborative learning environments, particularly in the context of blockchain's inherent transparency.

The combination of AI techniques with blockchain's immutable ledger establishes a comprehensive security framework for IoT transaction systems, enhancing data integrity and confidentiality. This integration not only automates and optimizes security protocols but also enables adaptive responses to emerging cyber threats, thereby increasing the resilience of IoT networks against cyber-attacks. Additionally, AI can analyze large datasets from IoT devices to identify patterns and anomalies indicative of security breaches, while blockchain ensures that all data records remain tamper-proof, further reinforcing the reliability of AI-driven security measures. This synergy is crucial in addressing privacy concerns associated with sensitive personal data collected by IoT devices, as AI can facilitate the development of protocols that protect user anonymity without compromising system functionality. The interplay between AI and blockchain technologies signifies a paradigm shift in how transaction security is approached, particularly in environments characterized by rapid technological advancements and evolving threat landscapes.

Emerging solutions effectively address the inherent conflict between transaction transparency and privacy preservation by utilizing advanced noise mechanisms and imple-

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menting differential privacy techniques, which provide robust quantitative privacy guarantees. These approaches enhance sensitive data privacy in high-stakes decision-making contexts and facilitate compliance with both the Right-to-Privacy (RTP) and Right-to-Explanation (RTE) requirements, ensuring that models remain interpretable and accountable. Recent studies have demonstrated that integrating privacy-preserving methods can significantly impact the quality of generated explanations in machine learning models, highlighting the necessity of considering the interplay between privacy and explainability to achieve optimal outcomes in safety-critical applications. Such considerations are essential as they navigate the delicate balance between leveraging data for security while safeguarding individual rights and compliance with regulatory frameworks. These techniques balance analytical utility with information protection during data aggregation, preventing sensitive information leakage while maintaining detection efficacy. The architectural integration of federated learning with blockchain technology enables decentralized anomaly detection while mitigating privacy risks associated with mutual information leakage during model training, establishing a comprehensive security framework that addresses both current vulnerabilities and emerging threats in blockchain-enabled IoT transaction systems.

Feature	System and network layer security	Smart contract and application layer security	On-chain data and transaction security
Security Focus	Intrusion Detection	Vulnerability Detection	Anomaly Detection
AI Technique	Federated Learning	Graph Neural Networks	Machine Learning
Blockchain Integration	Immutable Audit Trails	Automated Code Auditing	Immutable Ledger

表 3: This table provides a comparative analysis of AI-driven security methods across different layers within blockchain-enabled systems. It highlights the focus areas of security, AI techniques employed, and the integration of blockchain technology to enhance security measures. The comparison underscores the adaptability and potential of AI and blockchain synergy in addressing emerging threats and vulnerabilities in decentralized infrastructures.

4 Key Enablers and Specialized Architectures

Understanding blockchain's role in IoT requires examining specific enablers that facilitate its effective deployment, particularly lightweight blockchain designs and consensus mechanisms tailored for resource-constrained environments. These innovations address the limitations of traditional blockchain architectures, enhancing security and efficiency while enabling new applications and services that leverage the unique characteristics of blockchain and IoT technologies. By exploring these enablers, researchers and practitioners can better navigate the complexities of integrating blockchain within IoT ecosystems.

4.1 Lightweight Blockchain Design and Consensus Mechanisms for IoT

Standard blockchain architectures face challenges in IoT ecosystems due to their resource-intensive consensus mechanisms and storage requirements, which create computational bottlenecks in constrained environments [3]. Traditional Proof-of-Work (PoW) implementations demand substantial processing power and energy consumption, making them impractical for real-world IoT applications. Differential privacy implementations must balance security with computational feasibility, highlighting the need for specialized lightweight solutions that maintain blockchain's security guarantees while operating within strict resource constraints. These limitations can hinder the scalability and adoption of blockchain in IoT, necessitating innovative approaches to reconcile the trade-offs between security and efficiency.

Innovative architectural approaches address these limitations through optimized computation frameworks and decentralized trust mechanisms. The integration of Deep Sparse AutoEncoders for feature extraction demonstrates how lightweight machine learning techniques can significantly reduce computational overhead while maintaining detection accuracy in resource-constrained environments [9]. These solutions are effective in edge computing scenarios where traditional centralized approaches fail to accommodate device constraints, addressing energy inefficiencies of conventional consensus protocols through optimized algorithms tailored for IoT networks. Such lightweight designs facilitate the seamless integration of IoT devices into existing blockchain frameworks, enhancing interoperability and promoting a cohesive ecosystem. These architectural innovations are pivotal for improving performance and advancing IoT system functionality.

Privacy-preserving computation techniques enable practical blockchain deployment across heterogeneous IoT networks. A two-step process involving data pre-processing with deep learning architectures followed by ensemble classification exemplifies how computational efficiency can be achieved without compromising security [9]. These mechanisms operate in tandem with secure multi-party computation protocols that enable collaborative analytics without revealing individual data points, demonstrating improvements in efficiency and scalability compared to traditional blockchain approaches. Maintaining privacy while ensuring data integrity is critical in sensitive applications such as healthcare and finance, where data breaches have high stakes. Developing these privacy-preserving techniques fosters trust among users and stakeholders in IoT environments.

Innovations in consensus mechanisms designed for IoT environments have led to advancements in performance and security, crucial given IoT's unique challenges such as resource constraints and device heterogeneity. Integrating machine learning with blockchain technology automates and optimizes security protocols, enabling adaptive

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responses to emerging cyber threats. Anomaly detection systems within these frameworks enhance the capability to monitor and secure IoT networks, ensuring data integrity and privacy while mitigating vulnerabilities associated with traditional security measures [17, 16, 4, 45, 29]. Lightweight solutions leverage optimized cryptographic methods and hierarchical trust distribution to reduce computational and storage demands while maintaining security guarantees. Deep learning techniques combined with feature selection algorithms integrated with blockchain bolster security and data integrity in resource-constrained environments. These architectural innovations establish a foundation for practical blockchain deployment in IoT networks, enabling the integration of AI-driven security solutions while preserving decentralization and cryptographic trust.

The effectiveness of these lightweight solutions lies in their capacity to eliminate single points of failure, bolstering system resilience and fostering trust through enhanced transparency, as demonstrated by privacy-preserving techniques and explainable AI in high-stakes decision-making frameworks. These approaches ensure compliance with Right-to-Privacy (RTP) and Right-to-Explanation (RTE) standards and facilitate vulnerability detection in smart contracts, contributing to safer digital environments [46, 24, 47, 48]. Computational overhead challenges are addressed through innovative architectures maintaining robust security analysis capabilities while operating efficiently in resource-constrained environments. These specialized implementations demonstrate how blockchain's core properties can be optimized for IoT deployments, creating a foundation for advanced security frameworks addressing current vulnerabilities and emerging threats in decentralized IoT ecosystems.

The integration of lightweight blockchain designs and consensus mechanisms enhances network security and efficiency in IoT settings. As illustrated in Figure 4, two distinct architectures leverage blockchain technology to address common challenges in IoT systems. The "Disconnected IoT Subnet" example showcases a system designed to detect and predict attacks within an IoT subnet, featuring a Reputation System and a Predict Attack module, emphasizing proactive security posture and strategic data flow management to anticipate and mitigate threats. The "Blockchain-based IoT network architecture" example highlights a decentralized network structure with interconnected nodes, including a Distributed Hash Table (DHT) and an Edge node, where blockchain serves as a secure, decentralized platform for data storage and management, ensuring efficient data retrieval and robust security measures. These examples underscore the potential of specialized architectures and key enablers in revolutionizing IoT systems through innovative blockchain applications.

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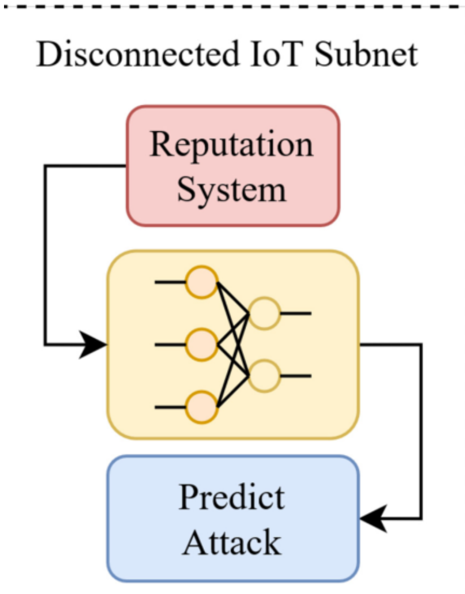
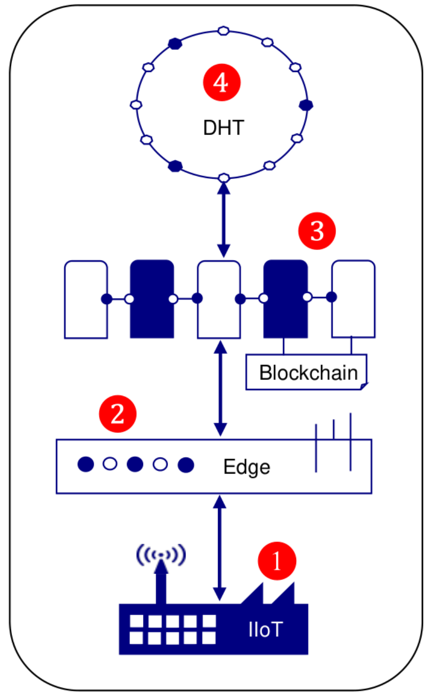


(a) Disconnected IoT Subnet[49]



(b) Blockchain-based IoT network architecture[17]

图 4: Examples of Lightweight Blockchain Design and Consensus Mechanisms for IoT

4.2 Federated Learning and Other Privacy-Preserving Machine Learning Methods for Decentralized Data

Federated learning (FL) has emerged as a foundational paradigm for privacy-preserving analytics in blockchain-enabled IoT networks, addressing data silo challenges through decentralized model training while maintaining strict data locality. The eFedDNN framework integrates federated learning with an ensemble of deep neural networks, demonstrating enhanced accuracy while preserving data privacy through distributed model aggregation [10]. This architecture proves effective in IoT environments where raw data sharing poses privacy risks, as FL enables collaborative model training without direct data exchange between devices [2]. Leveraging distributed data sources while ensuring privacy fosters trust among users and stakeholders, particularly in sensitive domains such as healthcare and finance, where data confidentiality is crucial.

Advanced privacy-preserving computation techniques complement federated learning by providing cryptographic guarantees for decentralized data processing. The SWIFT framework enables privacy-preserving machine learning in IoT networks through secure multi-party computation (MPC), converting ML programs into MPC-specific intermediate representations for secure execution across distributed nodes [31]. SecretFlow-SPU enhances this paradigm by implementing privacy-preserving ML operations through

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specialized secure processing units, optimizing performance while maintaining cryptographic security guarantees [32]. These techniques address limitations of traditional centralized approaches by eliminating single points of failure while preserving data confidentiality through decentralized computation. Integrating these mechanisms enhances machine learning applications' robustness and contributes to a resilient and trustworthy IoT ecosystem.

Combining federated learning and lightweight blockchain architectures establishes secure and verifiable execution environments facilitating privacy-preserving operations, addressing IoT device constraints. This integration enables multiple parties to collaboratively train machine learning models without sharing raw data, while implementing smart contract-based policies to ensure training process integrity and auditability. Consequently, this approach enhances model aggregation reliability and protects user data in applications like smart homes and healthcare systems [50, 51]. Blockchain's immutable ledger provides an audit trail for model updates and aggregation processes, enabling transparent verification of federated learning operations while maintaining data privacy. This synergy is valuable in healthcare IoT applications, where federated learning and blockchain enable secure collaborative analytics on sensitive medical data without centralized data collection. The architectural integration establishes comprehensive solutions for secure data management in IoT networks, addressing data silo and computational constraint challenges while maintaining robust privacy guarantees through cryptographic techniques and decentralized trust mechanisms.

Emerging research focuses on optimizing performance-efficiency trade-offs in privacy-preserving federated learning for resource-constrained IoT devices. Recent advancements in FL encompass adaptive model compression techniques and dynamic participant selection algorithms, significantly mitigating communication overhead while preserving model accuracy. These developments are crucial in heterogeneous device environments, where optimizing device participation is essential for efficient model convergence and maintaining high performance. New client selection strategies have demonstrated improvements in model coverage and accuracy, achieving faster data summary calculations and clustering times in large-scale FL settings. Privacy-preserving methods, such as Round-Robin based collaborative training, facilitate model training without sensitive data sharing, enhancing machine learning models' robustness and applicability across diverse user populations [52, 53, 54]. These advancements demonstrate how federated learning architectures can evolve to meet blockchain-enabled IoT networks' stringent requirements, balancing privacy preservation with practical deployment considerations across heterogeneous device ecosystems. Continued development of these techniques will play a critical role in enabling secure, decentralized AI applications while addressing privacy and efficiency challenges in distributed IoT environments.

5 Performance, Trade-offs, and Core Challenges

Evaluating AI-driven security solutions in blockchain-enabled IoT networks requires a comprehensive framework that encompasses key performance indicators. This framework is critical for understanding the interaction between detection accuracy, computational efficiency, and privacy preservation, particularly in safety-critical applications that adhere to regulatory and ethical standards. By incorporating privacy-preserving machine learning (PPML) and explainable AI (XAI), it provides insights into how privacy techniques like differential privacy affect model explanations and performance, helping practitioners optimize utility-privacy trade-offs while ensuring transparency and accountability [47, 46, 55, 56]. The following analysis explores specific performance indicators that define the effectiveness of security implementations in this context.

5.1 Analysis of key performance indicators

Benchmark	Size	Domain	Task Format	Metric
FLPPML[57]	10,280	Language Recognition	Language Prediction	Training Speed, Communication Cost
PPML-Bench[55]	70,000	Image Classification	Multi-Class Classification	F1-score, AUC
IEB[58]	10,000	Image Encryption	Attack Simulation	Accuracy, Recovery Rate
L2R-PMML[56]	124,313	Text Classification	Image Classification	Attacker Advantage
DPML[59]	200,000	Image Classification	Multi-Class Classification	Accuracy Loss, Privacy Leakage
PPML-TS[60]	40,000	Time-Series Classification	Classification	F1-score
PPML-EDA[61]	1,000	Data Privacy	Exploratory Data Analysis	Privacy Loss, Accuracy
FedEval[62]	1,000,000	Machine Learning	Model Evaluation	Accuracy, Communication Round

表 4: This table presents a comprehensive overview of representative benchmarks used in assessing privacy-preserving machine learning methodologies. It details various benchmarks across different domains, such as language recognition, image classification, and data privacy, highlighting their respective task formats and evaluation metrics. These benchmarks are crucial for understanding the performance and privacy trade-offs in AI-driven security solutions within blockchain-enabled IoT networks.

A multi-dimensional assessment framework is essential for evaluating AI-driven security solutions in blockchain-enabled IoT networks, with classification accuracy as the primary metric for model efficacy. Classification accuracy indicates a security model's effectiveness in threat detection by quantifying true results among total cases examined. The FL-IDS method shows competitive performance through accuracy, precision, recall, and F1-score measurements, achieving results comparable to centralized models while maintaining data privacy in federated learning environments [2]. These metrics are crucial for assessing detection capabilities, as demonstrated by the PRO-DLBIDCPS

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technique, which uses an Adaptive Harmony Search Algorithm for feature selection and attention-based Bi-directional Gated Recurrent Neural Networks for anomaly detection [5]. Various methodologies highlight the importance of enhancing detection efficacy while ensuring compliance with privacy standards.

Precision-recall trade-offs are significant in IoT security contexts, where DNN-Blockchain integration achieves 99.18% threat detection accuracy with a 15.42% false-positive rate, outperforming traditional systems [3]. High detection accuracy is critical in environments requiring timely threat identification. The F1-score's harmonic mean calculation is essential for addressing class imbalance in intrusion detection, providing a balanced evaluation of false positives and negatives. Metrics like the Matthews Correlation Coefficient (MCC) offer robust performance assessment across imbalanced datasets, as shown in evaluations of deep learning architectures for IoT anomaly detection [9]. These diverse metrics ensure a comprehensive understanding of model performance, guiding informed decisions about security solution deployment.

Computational efficiency metrics assess security solutions' viability in resource-constrained environments. Reaction time efficiency and scalability measurements reveal critical performance characteristics when comparing AI-enhanced approaches with conventional systems [3]. Statistical significance tests validate detection methods' robustness, particularly regarding feature selection algorithms' impact on model performance [9]. Table 4 provides a detailed overview of representative benchmarks that are instrumental in evaluating the efficacy of privacy-preserving machine learning techniques in AI-driven security solutions. These evaluations establish standardized benchmarks for comparing diverse security solutions while considering blockchain-enabled IoT ecosystems' unique constraints. Analyzing performance indicators for AI security solutions facilitates a thorough evaluation of their effectiveness, highlighting trade-offs among detection accuracy, computational efficiency, and privacy preservation. This analysis is crucial in safety-critical applications, where integrating privacy-preserving techniques and explainable AI methods is necessary for balancing privacy and explanation rights. Examining these interactions helps understand real-world applications' implications, guiding robust AI systems' development that meets regulatory and ethical standards while maintaining user trust [56, 11, 47, 46, 24]. These metrics inform decentralized security implementations' optimization, providing empirical foundations for evaluating solution effectiveness in practical IoT deployments. Comparative performance analysis across different architectural approaches establishes critical baselines for future research in AI-driven blockchain-IoT security frameworks.

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5.2 Discussion of key trade-offs

Implementing AI-driven security solutions in blockchain-enabled IoT networks involves trade-offs between computational efficiency, detection accuracy, and privacy preservation, each imposing constraints on resource-constrained deployments. Computational overhead is a primary challenge, especially in healthcare IoT applications where the FIDANN framework depends on variable bandwidth and computational capabilities across edge devices [7]. These limitations are evident in real-time security applications where privacy-preserving implementations must balance model complexity against resource constraints, creating tensions between high accuracy and deployment feasibility. Optimizing these trade-offs is essential for ensuring security solutions' effectiveness in diverse operational contexts while adhering to privacy regulations.

Accuracy-efficiency trade-offs persist across AI security implementations, with analyses revealing conflicts between detection performance and computational requirements. Integrating deep neural networks with blockchain technology demonstrates superior threat detection capabilities (99.18% accuracy) with manageable false-positive rates (15.42%) but at a significant computational cost [3]. These trade-offs are pronounced in federated learning environments where the FL-IDS framework achieves competitive detection performance while preserving data privacy through distributed model training [2]. Challenges extend to feature selection processes, where the PRO-DLBIDCPS technique uses an Adaptive Harmony Search Algorithm to optimize computational efficiency without compromising detection accuracy [5]. Addressing these trade-offs is vital for developing scalable security solutions that adapt to the evolving threat landscape.

Privacy-performance trade-offs impact architectural decisions in blockchain-IoT systems, where blockchain's immutable nature conflicts with privacy-preserving requirements. Lightweight blockchain adaptations address these tensions through optimized cryptographic methods and hierarchical trust distribution, enabling efficient AI security solutions deployment while maintaining privacy guarantees. The federated learning paradigm exemplifies this balance, allowing collaborative model training across IoT devices without centralized data collection [2], though with increased communication overhead and potential model convergence challenges. Navigating these trade-offs is crucial for designing effective security frameworks that operate efficiently within blockchain technology constraints while ensuring user data protection.

These trade-offs influence decentralized IoT ecosystems' security frameworks' design. Resource constraints necessitate innovative approaches that transcend limitations, such as adaptive model compression techniques and dynamic participant selection algorithms in federated learning environments. Solutions must reconcile transparency needs in blockchain technology with the imperative to protect sensitive data processed by IoT devices. This reconciliation is crucial for ensuring practical deployment across diverse device ecosystems, as blockchain and AI integration in IoT environments enhances data integrity and confidentiality while facilitating innovative privacy protocols development. These protocols safeguard user anonymity while maintaining IoT systems' functionality and security, addressing interconnected devices' escalating cybersecurity challenges [16, 29, 63, 17]. Exploring these trade-offs will inform future research, guiding more effective and adaptable security solutions development.

5.3 Core challenges faced by the field

Integrating AI with blockchain-enabled IoT networks faces challenges in adversarial resilience, data scarcity, and model interpretability, each presenting barriers to practical deployment in resource-constrained ecosystems. Adversarial attacks are increasingly sophisticated, with intrusion detection tools showing high false positive rates and limited emerging vulnerability detection [13]. Blockchain's immutability exacerbates these vulnerabilities by permanently recording compromised transactions, while federated learning schemes struggle to verify aggregated gradient correctness without compromising participant privacy [2]. Adversarial machine learning discussions highlight the need for unified privacy metrics across different models, a significant unresolved issue in the field [11]. Addressing these challenges is essential for developing robust security frameworks that withstand evolving threats.

Data constraints, such as non-IID distributions and heterogeneous device capabilities, degrade model accuracy in decentralized environments. Vertical federated learning frameworks face performance limitations dependent on overlapping samples' quality and quantity [33], while client scheduling methods demonstrate inefficiencies in model convergence due to outdated local updates. Struggles in detecting novel attacks not present in training data compound data scarcity challenges [64]. Communication overhead and node heterogeneity present additional barriers to efficient collaborative learning in IoT environments [65], with resource-intensive scenarios problematic for blockchain adaptation to diverse IoT devices [14]. Addressing these data-related challenges is crucial for enhancing AI-driven security solutions' reliability and accuracy.

Interpretability limitations hinder trust in AI-driven security decisions, particularly in complex models for vulnerability detection. Structured mappings of user acceptance criteria can enhance trust [66], but comprehensive solutions for explaining security-critical decisions in decentralized environments are lacking. These limitations are exacerbated by fully homomorphic encryption approaches' computational intensity when handling non-linear activation functions essential for complex models [67]. Balancing effective unlearning and model utility presents additional interpretability challenges, with over-unlearning and under-unlearning issues requiring careful consideration [68].

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Overcoming these challenges is essential for fostering user confidence in AI security systems.

Practical deployment challenges further constrain AI security solutions' adoption in blockchain-IoT networks. High computational costs and lack of open-source implementations hinder widespread adoption and research reproducibility [69]. Proposed methods that reduce resource consumption may still face challenges in high transaction volumes or variable network conditions [1]. Privacy preservation is a critical concern, with existing methods often prioritizing prediction accuracy at the expense of user privacy, potentially leading to data breaches and regulatory non-compliance [10]. Addressing these practical challenges is vital for successfully integrating AI and blockchain technologies in IoT applications.

Emerging solutions show progress through optimized federated architectures and lightweight secure computation implementations. The field increasingly adopts hybrid methodologies integrating AI and blockchain technology to enhance detection accuracy, optimize computational efficiency, and safeguard privacy. These approaches ensure necessary transparency for security-critical applications within the blockchain-IoT ecosystem, addressing sensitive data management complexities and responding to evolving cyber threats while maintaining interconnected devices' integrity and confidentiality [16, 63, 17]. Ongoing research focuses on developing comprehensive testing methodologies addressing smart contracts' immutability challenges while improving evolving vulnerability types' coverage. These advancements aim to overcome current approaches' limitations in resource-constrained environments, establishing robust frameworks for AI-driven security in decentralized IoT networks.

6 Future Research Directions

Exploring future security mechanisms in blockchain-enabled IoT networks necessitates innovative approaches to enhance adaptive responses to emerging threats. The proliferation of IoT devices has heightened vulnerability to cyber threats, demanding sophisticated security measures. Blockchain integration offers advantages like enhanced transparency and immutability, which can be leveraged to improve security protocols. This section examines the development and implementation of intelligent and automated security response systems, emphasizing their potential to revolutionize security practices through real-time threat detection and autonomous mitigation strategies. These systems must adapt to dynamic environments, ensuring resilience against evolving threats.

6.1 Intelligent and Automated Security Response Systems

Developing intelligent security response systems in blockchain-enabled IoT networks requires architectures that integrate real-time threat detection with autonomous miti-

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gation, addressing computational constraints and privacy preservation. Reinforcement learning offers transformative potential for automating security responses, providing adaptive countermeasures against sophisticated attacks while maintaining explainability in critical IoT environments [46]. These systems should incorporate dynamic analysis techniques to enhance detection capabilities and reduce false positives in smart contract vulnerability identification [34], optimizing the trade-off between security efficacy and computational efficiency through quantization-aware federated learning frameworks [70].

Integrating blockchain technology with collaborative anomaly detection systems presents opportunities for developing lightweight machine learning models suitable for resource-constrained IoT devices [4]. Future research should extend anomaly detection methods to handle multi-class security problems through advanced computational frameworks, enhancing model resilience against adversarial attacks in decentralized IoT environments [9]. The SWIFT framework shows promise for supporting complex machine learning tasks in security automation, particularly through secure multi-party computation protocols [31].

The intersection of privacy and fairness in security automation requires investigation, especially in federated learning environments where differential privacy impacts both model accuracy and equitable decision-making [71]. Vertical federated learning is promising for industrial IoT security automation, enabling collaborative threat detection across organizational boundaries while preserving data sovereignty [72]. The Version Age of Information (VAoI) metric offers novel opportunities for optimizing client scheduling in federated learning-based security systems, potentially improving model convergence and threat response times in heterogeneous IoT networks [73].

Privacy-preserving computation forms the foundation for decentralized security automation, with distributed privacy-preserving prediction (DPPP) methods combining differential privacy techniques and cryptographic protocols to safeguard individual contributions while maintaining prediction accuracy [74]. Future research must optimize these frameworks for IoT environments by extending random permutation techniques to diverse privacy-preserving machine learning models under varying security settings [75]. Comprehensive empirical plans to investigate the interplay between local differential privacy and adversarial robustness will be critical for advancing intelligent security response systems [76].

AI and blockchain convergence enables novel security automation paradigms through decentralized verification and immutable response mechanisms. Machine learning enhances security responses by automating decision-making for blockchain consensus mechanisms, demonstrating potential for adaptive threat mitigation in IoT networks [45]. Future research should focus on improving traceback effectiveness against sophisticated poisoning attacks in federated learning environments [43], while optimizing multiparty computation protocols for scenarios with fewer participants to enhance applicability in resource-constrained IoT deployments [77].

Emerging research must address reproducibility challenges in privacy-preserving techniques, with homomorphic encryption (HE) and secure multi-party computation (SMPC) showing promise for secure machine learning applications in IoT security automation [69]. Structured approaches to PPML technique selection, incorporating user acceptance criteria into decision-making processes, will advance security automation frameworks [78]. These directions collectively advance the field toward autonomous security systems capable of real-time threat detection, adaptive response, and verifiable execution in decentralized IoT networks.

6.2 The Role of Large Language Models (LLMs) in Blockchain Security

Large Language Models (LLMs) are transforming blockchain security through advanced automated code analysis and intelligent security management interfaces, addressing challenges in smart contract auditing and decentralized threat detection. The GPTLENS framework demonstrates superior vulnerability detection capabilities, leveraging LLMs' semantic understanding to analyze both natural language documentation and bytecode representations across diverse blockchain platforms [27]. This approach combines neural networks' capacity for complex code semantics with expert patterns, enhancing detection accuracy and explainability in security audits [40].

Integration of retrieval-augmented generation techniques with LLMs establishes new paradigms for smart contract security, combining model-generated insights with external knowledge bases to improve vulnerability detection accuracy [24]. These advancements enable continuous adaptation to new vulnerabilities while maintaining expert-level precision. The Smart Policy Control framework exemplifies LLMs' enhancement of federated learning security through verifiable and auditable management systems, ensuring both local and global model integrity in decentralized IoT networks [51]. This integration is particularly valuable for resource-constrained environments, demonstrating LLMs' potential to transform blockchain-IoT security architectures.

Natural language interfaces powered by LLMs enable intuitive security management across decentralized networks, bridging the gap between complex technical implementations and operational requirements. Future developments will focus on extending symbolic execution capabilities like those in WANA to enhance LLM-based vulnerability detection, particularly for WebAssembly bytecode analysis and external function handling [41]. These interfaces simplify security operations in heterogeneous IoT ecosystems while maintaining precision for critical security decisions. Reconciliation of privacy and explainability in high-stakes security contexts remains a challenge for

LLM implementations, necessitating continued research into interpretable model architectures [46].

Federated learning architectures present promising avenues for integrating LLMs with blockchain security frameworks while preserving data privacy. Future research directions include exploring split-learning settings and integrating Graph Convolutional Networks to enhance spatial-temporal data capture in security-related analytics [10]. These advancements will enable LLMs to operate effectively in decentralized environments, addressing challenges of data heterogeneity and device constraints in IoT networks. Development of comprehensive benchmarks for LLM-based security tools will advance the field, enabling continuous evaluation against evolving threats and establishing standardized performance metrics [28].

LLMs' transformative impact extends to automated response systems and adaptive threat mitigation in blockchain-IoT ecosystems. Emerging research must address limitations in current applications, particularly in optimizing model architectures for resource-constrained devices and enhancing method adaptability to new vulnerability patterns. Future directions should focus on integrating domain-specific knowledge into LLM-based security frameworks [39], while exploring novel applications in cryptographic algorithm optimization and privacy-preserving computation [79]. These developments will establish LLMs as foundational components of next-generation security systems, capable of addressing vulnerabilities and emerging threats through continuous learning and adaptation in decentralized IoT environments.

6.3 Opportunities for Fully Decentralized Autonomous Security Frameworks

The evolution of decentralized autonomous organizations (DAOs) integrated with AI-driven security mechanisms establishes paradigms for adaptive threat response in blockchain-enabled IoT networks, combining community governance with intelligent automation. Future research should prioritize developing lightweight, scalable security protocols capable of efficient resource management across heterogeneous IoT networks while maintaining robust protection against emerging vulnerabilities [6]. This integration necessitates advancements in privacy-preserving computation techniques, particularly for industrial IoT environments where context-aware security mechanisms must balance detection accuracy with computational constraints [80].

Proof-of-Honesty (PoH) consensus mechanisms exemplify decentralized security governance potential, combining cryptographic verification with behavioral trust metrics to establish resilient frameworks for IoT networks [45]. These architectures benefit from vertical federated knowledge transfer methods that enable cross-domain security intelligence sharing while preserving data sovereignty through decentralized learning

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paradigms [81]. The UA-PDFL framework's personalized approach to decentralized federated learning enhances communication efficiency and adaptive learning capabilities within autonomous security ecosystems [82].

Lightweight consensus mechanisms require optimization to support diverse IoT applications, with research focusing on enhanced scalability through innovative data management strategies and energy-efficient validation protocols [1]. Integrating these mechanisms with AI-driven anomaly detection creates comprehensive security frameworks capable of real-time threat response while maintaining decentralized governance principles. Future developments should explore the interplay between privacy preservation and algorithmic fairness in autonomous security systems, developing mitigation strategies addressing technical and sociotechnical implications of decentralized decision-making [83].

Communication-friendly privacy-preserving machine learning protocols are critical enablers for decentralized security frameworks, particularly when extended to support complex models in resource-constrained IoT environments [84]. These techniques must evolve to accommodate industrial IoT networks' unique requirements, where fully homomorphic encryption methods enhance secure data sharing while maintaining computational feasibility [85]. The convergence of these technologies establishes a foundation for community-driven security evolution, enabling continuous adaptation to emerging threats through transparent and verifiable mechanisms.

Developing standardized evaluation protocols and interoperability frameworks will be critical for advancing autonomous security ecosystems, ensuring security efficacy and practical deployment feasibility across heterogeneous blockchain-IoT applications. Future research should focus on optimizing adaptive machine learning integration with decentralized decision-making processes, addressing challenges in real-time threat response and resource-constrained operation. These advancements will establish next-generation security paradigms that integrate blockchain governance's decentralized and immutable nature with AI-driven analytics' adaptive capabilities, resulting in robust frameworks designed to enhance data integrity and confidentiality in IoT's rapidly evolving security landscape. By leveraging AI to analyze vast datasets and detect anomalies, alongside blockchain's secure management of device identities and transactions, these paradigms will automate and optimize security protocols, providing resilient defenses against sophisticated cyber threats targeting critical infrastructures. Additionally, this synergy will facilitate privacy-preserving protocol development, ensuring sensitive personal data collected by IoT devices remains protected without compromising system functionality [16, 17].

7 Conclusion

7.1 Summarize the core contributions of this paper

This paper significantly advances the understanding of AI-driven security within blockchain-enabled IoT networks through four key contributions. The first is the development of an extensive taxonomy that categorizes security solutions by architectural layers, incorporating technologies ranging from lightweight intrusion detection systems to advanced federated learning optimizations. This taxonomy highlights emerging methods, such as vertical federated learning, which are particularly beneficial for industrial applications due to their ability to maintain accuracy while minimizing computational demands. Additionally, it underscores privacy-preserving machine learning frameworks that balance security needs with IoT's computational limitations, thereby clarifying the security landscape and identifying future research opportunities.

The second contribution involves a detailed analysis of the performance trade-offs and implementation challenges associated with AI-blockchain integration. The study reveals that hybrid AI-blockchain models enhance intrusion detection precision while reducing authentication delays. It explores the tensions between privacy preservation and model utility, especially in IoT contexts, and examines promising privacy-enhancing techniques like differential privacy and secure multi-party computation. This analysis extends to healthcare applications, demonstrating the superiority of privacy-preserving medical image analysis over traditional methods, thus contributing to a nuanced understanding of implementing AI-driven security in sensitive areas.

The third contribution focuses on advancing specialized architectures tailored for IoT's unique constraints. This includes lightweight blockchain designs and consensus mechanisms optimized for resource-limited devices, as well as federated learning frameworks that address data silos and privacy concerns in decentralized networks. These architectural innovations illustrate how blockchain's core properties can be optimized to protect against evolving threats while accommodating IoT's limited resources, paving the way for future innovations that could further enhance IoT network security.

The fourth contribution identifies promising future research directions, such as developing intelligent automated security response systems, integrating large language models for smart contract auditing, and creating fully decentralized autonomous security frameworks that combine Decentralized Autonomous Organizations with AI-driven threat detection. Addressing gaps in adversarial resilience, cross-platform interoperability, and deployment feasibility is crucial for overcoming current adoption barriers. This forward-looking perspective provides a roadmap for scholars and practitioners to explore innovative solutions that enhance the security of blockchain-enabled IoT networks.

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Collectively, these contributions establish the integration of AI, blockchain, and IoT as a transformative paradigm for addressing fundamental security challenges. By demonstrating how adaptive threat detection, verifiable trust mechanisms, and efficient resource utilization can harmonize the conflicting requirements of decentralized ecosystems, the insights and frameworks presented in this review will guide the development of next-generation security frameworks in distributed environments, ultimately contributing to safer and more resilient IoT systems.

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